

AI-driven emotional intelligence in piano education: Deep learning models for expressive performance coaching

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ARTICLE INFO

Keywords:

AI-based music teaching tools
Learning feedback systems
Music education technology
Music performance analysis
Personalized feedback

ABSTRACT

With the increasing use of artificial intelligence (AI) in higher music education, there is a growing need to evaluate whether intelligent feedback systems can meaningfully enhance expressive piano performance. This study investigates the research question: *Does the use of an AI-based analytical performance assessment system lead to greater improvements in piano expressivity than traditional instruction?* To address this question, we developed an intelligent performance-assessment system that applies modern BiLSTM attention technologies to analyze expressive qualities of piano playing, including dynamics, phrasing, and emotional conveyance. A total of 290 Chinese undergraduate piano students (aged 18–22) were randomly assigned to an AI-supported group ($n = 145$) or a traditional-instruction control group ($n = 145$) for a three-month training period. Expressivity was measured using a hybrid 100-point scoring system that integrated objective performance analytics with expert double-blind evaluations. Statistical analysis included paired and independent t -tests and effect-size estimation. The intervention group demonstrated a substantial improvement in expressive performance ($\Delta M = 12.97$; $t(144) = -21.64$; $p < .001$; $d = 1.80$), indicating a large effect. By contrast, changes in the control group were minimal and statistically nonsignificant ($p = .092$; $d = 0.14$). These findings provide strong empirical evidence that AI-based feedback can significantly enhance students' emotional expressivity in piano performance.

1. Introduction

Music education is increasingly incorporating digital and artificial intelligence (AI) technologies, which are revolutionizing how performance training is approached (Sanganeria & Gala, 2024; Slavuj et al., 2024; Zhang, 2023). However, a persistent challenge in traditional piano pedagogy remains the imbalance between technical precision and emotional expressivity. While expressive communication—such as dynamics, phrasing, and emotional intention—is crucial to meaningful performance, students often receive instruction that places greater emphasis on technical correctness rather than emotional depth (Bonnaire & González-Moreno, 2022; Hu, 2024; Jia, 2023; Jiang, 2025). Research shows that this focus often results in performances that are technically accurate but lack emotional impact, mainly due to underdeveloped emotional intelligence and difficulties in expressing emotional intentions (Li, 2022; Liao, 2022; Spies, 2015; Wang, 2024).

At the same time, rapid advancements in AI and deep learning offer new opportunities to address this imbalance in pedagogy. Recent models built on recurrent neural networks, long short-term memory

architectures, and Transformer-based systems have shown impressive abilities to analyze sequential musical data, detect expressive variations, and identify micro-level dynamic and temporal features that traditional instruction might miss (Chen, 2025; Han, 2025; Lei & Liu, 2022; Liao, 2022; Peng, 2024; Rizal & Milyartini, 2024; Rui & Zhang, 2024; Tang, 2025; Zhai & Xu, 2023; Zhang et al., 2024). Despite these advancements, there are still gaps in the literature. Most AI-driven tools focus primarily on technical accuracy rather than expressive qualities; few systems combine automated analysis with expert evaluations; and there is a lack of empirical studies on whether AI can actually enhance emotional expressivity in musical performances (Huang & Ding, 2022; Ji & Tong, 2024; Slavuj et al., 2024; Wang, 2025; Yeh, 2016).

The AI-supported system examined in this study is theorized to operate through indirect psychological and behavioral mechanisms rather than through explicit instructional guidance. Specifically, its use is assumed to influence performance by enhancing performers' awareness of expressive features and supporting self-monitoring during practice and performance. By externalizing expressive performance indicators in a structured and quantifiable form, the system provides

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<https://doi.org/10.1016/j.actpsy.2026.106264>

Received 3 July 2025; Received in revised form 13 January 2026; Accepted 14 January 2026

Available online 19 January 2026

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performers with a stable reference for evaluating expressive outcomes, without prescribing corrective actions or interpretive strategies.

At the behavioral level, this enhanced outcome awareness is expected to affect how performers allocate attention during rehearsal, leading to more focused engagement with expressive dimensions of performance. Accordingly, the effects of the intervention are operationalized not as changes in instructional behavior, but as measurable changes in performance outcomes along analytically defined dimensions of expressivity. This framing establishes a direct conceptual link between the theoretical mechanism (enhanced performance awareness and monitoring), the form of AI support implemented, and the outcome indicators used to evaluate intervention effects.

1.1. Literature review

1.1.1. Emotional intelligence in musical performance

Emotional intelligence is generally understood as the ability to perceive, interpret, and manage one's own emotions, as well as those of others. In music education, this concept extends to a performer's ability to grasp the emotional content of a piece, shape expressive nuances through dynamics and phrasing, and convey emotional meaning to the audience (Jia, 2023; Jiang, 2025). Traditionally, developing these abilities has relied on one-on-one instruction, where teachers guide students through the emotional layers of music and model expressive techniques (Peng, 2023). However, this approach is often subjective, influenced by the teacher's artistic preferences, and as a result, the quality and consistency of expressive training can vary greatly. These limitations point to the need for more objective and transparent tools that can assess and support emotional expressivity as a measurable skill (Kim et al., 2022; Li & Wang, 2024).

1.1.2. Theoretical framework

The pedagogical rationale for integrating AI-based feedback into piano education can be understood through two complementary theoretical lenses: the social-cognitive model of self-regulated learning (Zimmerman & Schunk, 2011) and the flow theory of optimal experience (dos Santos et al., 2018). Together, these frameworks offer a coherent explanation of why structured, real-time, and quantitatively grounded feedback may enhance musical expressivity and interpretative development.

According to Zimmerman and Schunk (2011), self-regulated learning (SRL) is a cyclical process in which learners actively plan, monitor, and evaluate their performance. Successful performers engage in three interrelated phases: forethought (goal setting and strategic planning), performance control (self-monitoring and use of feedback), and self-reflection (self-evaluation and adaptive adjustments). Central to SRL is the learner's ability to interpret feedback accurately and translate it into actionable strategies. In this context, AI-generated indicators of expressive parameters—such as dynamic expressivity, temporal stability, or musical coherence—provide objective benchmarks that support the performance control and self-reflection phases. They reduce ambiguity in students' assessment of their own expressive choices and help them identify specific technical and interpretative adjustments. Thus, the AI-supported environment strengthens the core SRL mechanisms by offering clear performance cues, structured monitoring, and increased metacognitive awareness, all of which are essential for the deliberate practice of expressive skills.

Flow theory, as discussed by dos Santos et al. (2018), further explains how learners sustain engagement and motivation during complex artistic tasks. Flow occurs when the balance between task difficulty and learner skill is optimal, goals are clearly defined, and feedback is immediate and informative. Educational systems that support flow tend to promote deeper concentration, intrinsic motivation, and meaningful learning experiences. Applied to piano performance, AI-based feedback contributes to these preconditions by translating abstract expressive qualities into concrete, interpretable metrics. This process helps students

maintain clarity of goals (e.g., “increase dynamic contrast”), receive immediate information about their performance, and perceive progress in a measurable way. Dos Santos et al. (2018) emphasize that such environments facilitate immersion and active involvement, which are particularly important in artistic domains requiring sustained focus and embodied engagement.

Together, SRL and flow theory provide a theoretical foundation for the proposed hybrid human–AI model. While SRL explains how learners use feedback to refine expressive strategies, flow theory elucidates why the presence of clear goals and continuous feedback enhances the depth of engagement during musical practice. The convergence of these frameworks suggests that AI-supported analytics can reinforce self-regulated practice, increase interpretative awareness, and create learning conditions conducive to both artistic growth and high motivational involvement.

1.1.3. AI-based analysis of expressive features

Recent advances in artificial intelligence have sparked interest in computational methods for analyzing expressive musical performance. Deep learning architectures, such as recurrent neural networks, long short-term memory models, and BiLSTM structures, are now capable of detecting expressive shortcomings like insufficient dynamic contrast, rigid timing, or monotonous articulation (Han, 2025; Ji & Tong, 2024; Lei & Liu, 2022). These models, when trained on high-quality professional performances, can also compare student interpretations to established expressive norms, helping learners pick up on subtle emotional cues in phrasing and tone (Li, 2022; Liu & Guo, 2025; Zhang et al., 2024). Additionally, research exploring real-time visualizations of expressive parameters shows that algorithms capable of displaying fluctuations in dynamics and tempo can help students become more aware of how technical decisions affect expressive outcomes (Chen, 2025; Sangneria & Gala, 2024; Tang, 2025). More advanced multimodal systems integrate audio data with facial expressions, motion patterns, or physiological responses, offering a fuller picture of emotional conveyance during performance (Rizal & Milyartini, 2024; Wang & Iida, 2024; Yin & Guo, 2025). Together, these developments suggest that AI can offer consistent, data-driven support in developing expressive musical skills.

1.1.4. Gaps in existing research and implications for the present study

Despite these promising advancements, challenges still exist. One major hurdle is the need for large annotated datasets, which are currently lacking in the field of expressive piano performance (Cheng et al., 2025; Zhai & Xu, 2023). Existing AI tools often focus on technical accuracy—such as intonation, tempo consistency, or error detection—while emotional expressivity is still an underexplored area (Slavuj et al., 2024; Wang, 2025; Yeh, 2016). Additionally, few studies integrate automated computational results with expert-based evaluations, which limits the practical use of these systems in teaching. There is also limited empirical evidence showing that AI-driven feedback can significantly enhance students' expressive performance, particularly in culturally diverse learning environments (Ji & Tong, 2024; Peng, 2023; Zhang et al., 2024). As a result, there is a pressing need for AI-supported models that can analyze key expressive parameters—like dynamic expressivity, musical coherence, and emotional conveyance—and provide personalized feedback that addresses students' interpretative growth (Wang, 2022; Yin & Guo, 2025).

The present study seeks to fill this gap by evaluating whether AI-based feedback can improve emotional expressivity more effectively than traditional instruction. While the literature suggests that intelligent systems hold promise for enhancing expressive skills, both theoretical and empirical foundations remain underdeveloped. This highlights the importance of testing and refining an AI-based model, such as the BiLSTM attention-based system, which is designed to assess key expressive dimensions and complement expert judgment within a structured teaching framework.

1.2. Problem statement

Music education is currently facing the need to create a new approach to cultivating emotional intelligence in piano students using intelligent assessment and feedback systems. With the distance learning practices becoming more widespread, piano enthusiasts gained an increasing interest in automated piano-playing devices that prioritize emotional delivery (Peng, 2024). Therefore, this study aims to develop and empirically evaluate the effectiveness of an AI-driven deep learning-based model (DLM) tailored to dissect the expressive attributes of a piano performance. The objectives of the study are (1) to build a DLM model for analyzing individual cues to emotional expression in piano music; (2) to empirically assess the effectiveness of the proposed model, focusing on its practical benefits.

2. Methods

2.1. Participants

The study involved 290 piano majors from two music colleges in Beijing. The sample included 151 men and 139 women aged 18 to 22. All participants demonstrated comparable levels of prior preparation, as confirmed by a preliminary assessment of phrasing and basic expressiveness. Inclusion criteria consisted of the required age range, piano specialization, and the absence of auditory or motor impairments that could affect performance. All students provided written informed consent in accordance with the Declaration of Helsinki.

Participants were randomly assigned to two equal groups: an AI-supported feedback group ($n = 145$) and a traditional instruction group ($n = 145$). Randomization was conducted by an independent assistant using the RAND function in Excel. Group assignments were placed in sequentially numbered, opaque, sealed envelopes that were prepared in advance by a member of the research team not involved in data collection or instruction. Envelopes were opened only after participant enrollment was completed, ensuring that neither participants nor instructors could influence group allocation at the time of assignment. Instructors were informed of group assignment only after randomization had been finalized and solely for the purpose of implementing the assigned instructional condition. Importantly, instructors were not involved in outcome assessment, and evaluators remained blinded to group allocation, thereby minimizing the risk of allocation bias and expectancy effects. Prior to the intervention, no statistically significant differences were found between the groups on key baseline characteristics.

For subjective evaluation, an independent sample of 120 listeners (60 musicians and 60 non-musicians) was recruited, with no overlap with the main sample. A weighting factor of 1.5 was applied to musicians' ratings to reflect their higher sensitivity to performance nuances, consistent with expert-weighted evaluation practices. Inter-rater reliability was satisfactory ($ICC = 0.78$).

2.2. Materials

All performances were recorded on a Smart Piano digital instrument, which provides precise capture of MIDI parameters, including note onset time, duration, key velocity, and pedal activity. Prior to data collection, the instrument was calibrated to ensure stable parameter acquisition. The MIDI data were aligned with the reference scores using a hidden Markov model (HMM), which ensured valid temporal synchronization for subsequent expressive performance analysis.

The repertoire comprised two Chinese folk pieces ("A Merry Song of Flowers", "Flowers on the Field") and two excerpts from Western classical music: the first movement of Beethoven's Piano Sonata No. 14 in C-sharp minor, Op. 27 No. 2 ("Moonlight") and Chopin's Prelude Op. 28 No. 4. These works were selected due to their rich phrasing structures and their demands for finely nuanced dynamic shaping.

The intervention was based on a deep learning model that simultaneously processed MIDI and audio data. Audio was used exclusively to generate Mel spectrograms required for the analysis of timbral and textural structure. The model architecture is presented in Table 1. It consisted of a bidirectional LSTM block, an attention mechanism, a dropout layer, a fully connected layer, and an output layer with three regression neurons predicting dynamic expressivity, musical coherence, and emotional conveyance. The model was trained using the Adam optimizer ($lr = 0.001$), mean squared error (MSE) as the loss function, and 5-fold cross-validation with early stopping.

The model's output parameters served as quantitative indicators of expressiveness that paralleled the subjective evaluations. The expressiveness assessment system comprised a 40-point objective scale and a 60-point subjective scale. The objective indicators included rhythmic accuracy, dynamic expressivity, and temporal regularity, each standardized to a 0–10 range. The subjective scale encompassed three dimensions: dynamic expressivity, musical coherence, and emotional conveyance. The internal consistency of the subjective ratings was high (Cronbach's $\alpha = 0.89$).

To ensure terminological consistency and measurement transparency, a standardized mapping was established between listener-rated dimensions, AI model outputs, and the composite expressivity score. Although similar constructs are referenced across subjective evaluation, computational analysis, and integrated scoring, a single set of standardized dimension labels was adopted throughout the manuscript. Table 2 presents a one-to-one alignment of the three expressivity dimensions as assessed by listeners, operationalized by the BiLSTM-based model, and incorporated into the composite expressivity score.

Fig. 1 presents the conceptual model of the study, illustrating the alignment between the theoretical frameworks of Self-Regulated Learning and Flow Theory, the AI-supported analytical assessment environment, and the expressivity outcome dimensions. The model specifies how distinct psychological and behavioral mechanisms are operationalized through analytically defined performance indicators used to evaluate intervention effects. Each expressivity dimension used in the present study was explicitly mapped onto the underlying theoretical frameworks to ensure conceptual and measurement alignment. Within the Self-Regulated Learning framework, Dynamic Expressivity reflects performers' monitoring and regulation of intensity and dynamic range, whereas Musical Coherence captures goal alignment and structural organization across phrasing and timing. From the perspective of Flow Theory, Musical Coherence represents sustained attentional absorption and temporal continuity during performance, while Emotional Conveyance corresponds to emotionally integrated execution and experiential engagement. Musical Coherence thus serves as a theoretically overlapping outcome dimension informed by both self-regulatory and flow-related mechanisms. Together, these item-level correspondences establish a direct link between theory, operationalization, and outcome indicators used to evaluate intervention effects.

Table 1
Architecture of a deep learning model for expressive performance analysis.

Elements	Description	Parameters
Input	Data extracted from MIDI and audio files	Velocity, Pitch, Duration, MEL-spectrogram
BiLSTM	Bidirectional recurrent neural network (BRNN)	2×256 neurons
Attention block	Focuses on significant parts of the music data	Self-Attention Layer
Dropout	Prevents overfitting	Dropout = 0.3
Dense layer	A fully connected layer with a ReLU activation function	128 neurons
Output layer	Performs regression to predict the target values	3 outputs: Dynamics, Melodic sense, Emotional conveyance (0–5)

Table 2
Terminology and indicator mapping table.

Standardized Dimension (Used Throughout Manuscript)	Listener-Rated Dimension (Subjective Scale)	AI Model Output (BiLSTM Regression)	Contribution to Composite Expressivity Score
Dynamic Expressivity	Dynamic expressivity (listener rating)	Dynamic expressivity (velocity-based index)	Included in subjective (0–20) and objective (0–10) subscores
Musical Coherence	Musical coherence / musical coherence	Musical coherence (temporal–structural stability)	Included in subjective (0–20) and objective (0–10) subscores
Emotional Conveyance	Emotional conveyance / emotional grasp	Emotional conveyance (multimodal regression output)	Included in subjective (0–20) and objective (0–10) subscores

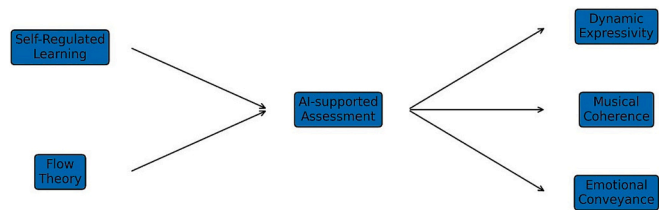


Fig. 1. Conceptual model linking theoretical frameworks, AI-supported assessment and expressivity outcome dimensions.

2.3. Procedure

The procedure was conducted over 13 weeks of the academic semester. During the first week, all participants produced a standardized recording of the repertoire under identical acoustic conditions and using the same equipment. Recordings were scheduled at the same time of day, and participants were instructed to refrain from practicing the repertoire for 24 h prior to testing to minimize short-term learning effects.

From Week 2 to Week 12, students attended weekly individual lessons of 45 min. The instructional program, teaching guidelines, and repertoire were identical for both groups; the only difference lay in the type of feedback provided. In the control group, the instructor used conventional verbal feedback, whereas in the experimental group part of the feedback was replaced with the model's numerical indicators. The model did not provide artistic suggestions, which ensured that the intervention focused solely on analytical performance metrics.

In Week 13, participants completed a post-test recording under the same conditions. The expert evaluation procedure was fully double-blind: listeners were unaware of both group assignment and recording stage.

All participants were instructed to maintain a consistent amount of individual practice. The absence of objective monitoring of practice duration is acknowledged as a methodological limitation.

2.4. Research design

The study employed a randomized experiment with two parallel groups. The independent variable was the type of feedback (AI-supported versus traditional), and the dependent variables were the three expressiveness subscales and the overall 100-point score. An a priori power analysis indicated that a sample of 290 participants provided sufficient power (> 0.99) to detect a large effect ($d = 0.80$) at $\alpha = 0.05$. Data analysis was conducted in SPSS v24. Normality and homogeneity

of variance were confirmed using Levene's test. Independent *t*-tests were used to assess group equivalence at pretest, paired *t*-tests to examine within-group changes, and independent *t*-tests with Cohen's *d* to evaluate between-group differences at posttest. The significance threshold was set at $p < .05$.

To ensure the internal validity of the intervention and to minimize potential instructor- and context-related confounds, several procedural controls were implemented in the study design. Piano instruction was delivered by a total of six instructors affiliated with the participating music colleges. All instructors held comparable academic qualifications and teaching experience in undergraduate piano education. Prior to the start of the intervention, instructors participated in a coordination meeting in which the study protocol, instructional objectives, repertoire, lesson structure, and assessment procedures were reviewed in detail.

Instructor assignment was organized to reduce teacher effects and cross-group contamination. Each instructor taught students in only one experimental condition (either the AI-supported assessment group or the traditional instruction group), thereby preventing instructors from simultaneously applying different instructional approaches across groups.

The instructional curriculum was standardized across groups with respect to repertoire, lesson duration, lesson frequency, and core pedagogical objectives. Instructors followed a shared lesson framework that specified technical focus areas, expressive goals, and rehearsal structure, while allowing for natural pedagogical flexibility consistent with higher music education practice. The sole systematic difference between groups concerned the presence or absence of AI-generated analytical performance indicators.

To support implementation fidelity, instructors received written guidelines describing the permitted and non-permitted forms of feedback. In particular, instructors in the AI-supported group were instructed to reference only the numerical performance indicators generated by the system and to avoid providing additional interpretative explanations derived from the model outputs. Instructors in the control group followed conventional verbal instruction practices without access to AI-generated indicators.

To enhance the psychological validity of measuring emotional conveyance, the present study employed a combined assessment system grounded in principles of convergent and content validity. First, the subjective component of the scale was conceptually aligned with established psychological and music-pedagogical frameworks of emotional conveyance, encompassing dimensions such as dynamic expressiveness, phrasing coherence, and emotional transmission, which are widely recognized in the literature as key behavioral markers of emotional communication in musical performance. Second, the high internal consistency of the subjective ratings (Cronbach's $\alpha = 0.89$) and satisfactory inter-rater reliability ($ICC = 0.78$) indicate measurement stability and consistency of interpretation across independent evaluators. Third, the psychological validity of the scale is further supported by the empirical convergence between subjective ratings and the model's objective parameters (dynamic variability, melodic coherence, and emotional engagement), suggesting that the assessment system captures reproducible behavioral manifestations of emotional conveyance rather than arbitrary aesthetic preferences.

2.5. Ethical consideration

The study was conducted in compliance with the Helsinki Declaration and applicable national and university regulations related to the conduct of research involving human subjects. All students received full information about the objectives and format of the intervention, including the AI-powered piano learning app. Each participant gave an informed, voluntary consent to participate in the research and to the processing of data obtained throughout the course of the intervention. The study follows the principle of confidentiality. Personal data were neither collected nor used in the analysis. All performance recordings

and data extracted from the assessment were anonymized and stored in an encrypted form with limited access. Participants had the right to withdraw from the research at any time without explanation and without any consequences. There was no forced interference with the student's academic trajectory. The research procedures were preliminarily reviewed and approved by the Ethics Committees of the participating educational institutions. The present findings are generalized for the purpose of assessing the effectiveness of the proposed instructional model.

2.6. Limitations

This study has several limitations that must be considered when interpreting the findings.

- (1) The study only included individuals who were pursuing piano education in Beijing. These were first- and second-year students, which limits the generalizability of the results to other age groups, educational contexts, music instruments, and novice pianists. In future research, it would be advisable to include learners of different age groups, cultural backgrounds, and levels of prior training to enhance external validity.
- (2) The musical repertoire selected for the intervention consists of technically accessible piano pieces chosen to master the basic elements of musical expression (i.e., dynamics, phrasing, and emotional tone). Hence, the study offers no insights into the effectiveness of the proposed model against more complex pieces with multi-layered texture, extravagant rhythms, or an expanded dynamic palette.
- (3) The DLM-based model considers a total of three expressive elements, but does not cover the entire set of expressive performance features, such as pedaling, articulation applied by the left and right hands, and intonation. Also, the attributes of an individual interpretive style, which may not coincide with the template embedded in the model's training algorithm, were not considered.
- (4) Note that the AI-driven model was evaluated in a controlled learning environment. Therefore, its effectiveness in a more flexible pedagogical context (e.g., self-study or improvisation) requires further investigation. These limitations open up new opportunities for future research, including expanding the sample, introducing more complex musical material, and developing algorithms that can take into account the multidimensional profile of musical expression.
- (5) A three-month instructional period may be insufficient to capture the long-term effects of the AI system on emotional expression. A longitudinal design is therefore recommended for a more comprehensive understanding of these dynamics. Finally, incorporating participants' subjective experiences and their reception of AI-generated feedback would provide a more complete account of its educational impact.

3. Results

Descriptive statistics for both groups at pre- and post-test are presented in Table 3. At baseline, the intervention and control groups demonstrated comparable levels of expressive performance. The intervention group had a mean expressivity score of 68.28 (SD = 5.20), while

the control group scored 67.59 (SD = 4.76). After the intervention, the intervention group showed a marked increase in expressivity (M = 81.24, SD = 4.98), whereas the control group demonstrated minimal change (M = 68.54, SD = 4.58).

An independent-samples t-test confirmed baseline equivalence between the groups. Variances did not differ significantly ($F = 0.707, p = .401$), allowing for a parametric comparison. The pre-test difference was small and statistically nonsignificant, $t(288) = 1.178, p = .240$, with a 95% CI for the mean difference of -0.462 to 1.842 , indicating no initial advantage for either group.

After the intervention, homogeneity of variances was again confirmed ($F = 0.469, p = .494$). The difference between groups became highly significant, $t(288) = 22.618, p < .001$. Crucially, the effect was also large in magnitude: the mean difference was 12.70 points (95% CI [11.60, 13.81]), reflecting a substantial practical improvement associated with AI-assisted training. Detailed statistics are presented in Table 4.

A paired-samples analysis showed that students in the intervention group experienced a strong improvement over time. Expressivity increased by 12.97 points on average (SD = 7.22), with a 95% CI of -14.15 to -11.78 . This change was statistically significant, $t(144) = -21.638, p < .001$, and represents a large effect, $d = 1.80$, indicating a robust and practically meaningful enhancement in expressive performance (Table 5). In contrast, the control group demonstrated only a minimal, statistically nonsignificant increase of 0.95 points (SD = 6.76), 95% CI $[-2.06, 0.16]$, $t(144) = -1.696, p = .092$. The effect size was small ($d = 0.14$), suggesting that traditional instruction did not produce meaningful gains in emotional expressivity.

The magnitude of changes in expressivity was quantified using the Cohen's d . For the intervention group, the value of Cohen's d is -1.80 , which represents a large effect size. This d value indicates a significant and practically meaningful improvement in expressivity following the completion of the AI-driven intervention program based on personalized feedback. For the control group, the value of Cohen's d is significantly lower (-0.14) and represents a small effect size. This result indicates an insignificant change in the level of expressivity. The comparison of the effect sizes across groups yielded evidence for the benefits of integrating AI technologies in music education and their positive impact on the expressive capabilities of pianists.

The weighting of listeners' ratings was introduced a priori to increase sensitivity to expert judgments in the evaluation of expressive performance, rather than to "balance" perspectives. A coefficient of 1.5 for musicians' ratings was selected based on two considerations: first, musicians demonstrated higher internal consistency (ICC = 0.82) compared to non-musicians (ICC = 0.71), and second, prior research indicates that trained listeners are more reliable in detecting micro-level expressive features such as dynamic shaping and phrasing coherence (Bonnaire & González-Moreno, 2022; Han, 2025; Kim et al., 2022; Li, 2022; Liao, 2022). To assess the robustness of the findings, a sensitivity analysis was conducted using unweighted ratings (i.e., equal contribution of musicians and non-musicians). The unweighted analysis yielded a slightly reduced but still substantial intervention effect. Specifically, the post-test between-group difference remained highly significant, $t(288) = 19.87, p < .001$, with a mean difference of 11.58 points (95% CI [10.44, 12.71]) compared to 12.70 points (95% CI [11.60, 13.81]) in the weighted model. The corresponding effect size decreased from $d = 1.80$ to $d = 1.64$, indicating a modest attenuation of magnitude without a change in statistical significance or direction of effects. Importantly, conclusions regarding the superiority of AI-supported feedback over traditional instruction remained unchanged across weighting schemes, confirming that the observed improvements in expressive performance are not driven by the weighting procedure but are robust to alternative evaluation assumptions.

Table 3
Distribution of mean values in the intervention and control groups.

Groups		Pretest	Posttest
Intervention	Mean	68.28	81.24
	N	145	145
	Std. Deviation	5.195	4.975
	Minimum	52	70
	Maximum	87	97
Control	Mean	67.59	68.54
	N	145	145
	Std. Deviation	4.763	4.581
	Minimum	55	57
	Maximum	80	79

Table 4
Results of the independent samples *t*-test.

		Pretest		Posttest	
		Equal variances assumed	Equal variances not assumed	Equal variances assumed	Equal variances not assumed
Levene's Test for Equality of Variances	F	0.707		0.469	
	Sig.	0.401		0.494	
t-Test for Equality of Means	t	1.178	1.178	22.618	22.618
	df	288	285.855	288	286.059
	Sig. (2-tailed)	0.240	0.240	0.000	0.000
	Mean Diff.	0.690	0.690	12.703	12.703
	Std. Error Diff.	0.585	0.585	0.562	0.562
	95% Confidence Interval of the Difference	Lower	−0.462	11.598	11.598
		Upper	1.842	13.809	13.809

Note: Mean differences reflect the direction of group comparisons as defined in the analysis. Effect sizes are reported with their original sign, which is determined by the order of comparison rather than by the substantive direction of the effect.

Table 5
Results of the paired *t*-test.

	Paired Differences				t	df	Sig. (2-tailed)	
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Intervention	−12.966	7.215	0.599	−14.150	−11.781	−21.638	144	0.000
Control	−0.952	6.758	0.561	−2.061	0.158	−1.696	144	0.092

Note. For paired comparisons, mean differences were calculated as pre-test minus post-test values; therefore, negative values indicate an increase in expressivity over time. The sign of *t* and Cohen's *d* follows the same calculation rule and reflects the order of subtraction rather than the substantive direction of change.

4. Discussion

The findings of this study indicate that the AI-driven DLM-based system substantially enhances students' emotional expressivity in piano performance. The improvement observed in the intervention group suggests that individualized analytic feedback supports the development of expressive features such as dynamic expressivity, phrasing, and emotional coherence. These results align with contemporary research showing that AI-assisted tools can meaningfully strengthen expressive and interpretative skills in instrumental education (Cheng et al., 2025; Kim et al., 2022; Tang, 2025).

A key point of convergence with earlier studies concerns the role of automated feedback in shaping students' understanding of expressive parameters. Prior works demonstrate that DL-based algorithms help learners notice subtle deviations in dynamics, timing, and articulation that may be difficult to detect independently (Han, 2025; Li, 2022; Liao, 2022). At the same time, some studies warn that AI tools may encourage overly mechanical or template-based performance if students rely exclusively on algorithmic instructions (Lei & Liu, 2022; Zhai & Xu, 2023). The present study does not reveal such risks, which may be explained by the hybrid instructional design.

The comparison between our results and existing literature also highlights an important distinction. Many previous systems prioritize technical accuracy as the primary outcome, while emotional expressivity is treated as an auxiliary parameter. By contrast, the present model was specifically designed to target expressive qualities, which may explain the stronger improvement in this domain. This focus is consistent with emerging trends advocating for AI-based tools that can interpret emotional intention and not merely measure performance precision (Rizal & Milyartini, 2024; Yin & Guo, 2025).

Because all participants were students of music colleges with many years of formal training, their baseline expressive performance skills were already highly developed. The repertoire used in the study consisted of relatively simple pieces selected to standardize the instructional conditions. Under such circumstances, substantial improvement among experienced performers may be constrained by a ceiling effect, which may account for the moderate changes observed in the control group.

Several mechanisms may account for the strong effect observed. First, the BiLSTM architecture with an attention mechanism likely contributed to the accurate detection of expressive cues spread across temporal sequences. Second, the integrated scoring system, which combined algorithmic outputs with weighted expert evaluations, provided students with multidimensional feedback that reflected both technical correctness and artistic coherence. Such dual-source assessment aligns with studies showing that multimodal feedback supports deeper interpretative understanding and fosters durable skill acquisition (Ji & Tong, 2024; Liu & Guo, 2025).

The results indicate that AI led to a more substantial improvement in performance expressiveness compared to traditional instruction. However, these findings should be interpreted with caution. The gains observed in the AI group may be attributed to the system's provision of quantitative benchmarks for key parameters—dynamic expressivity, musical coherence, and emotional conveyance. These indicators helped structure students' attention and enabled them to move from general pedagogical suggestions toward specific technical adjustments, aligning with prior evidence on the effectiveness of analytical technologies in music education (Han, 2025; Lei & Liu, 2022). At the same time, it is important to emphasize that the AI system did not replace artistic interpretation (Liao, 2022; Peng, 2023).

Sub-analyses provide further clarity on the nature of the observed improvements. The gender-based analysis showed that female students demonstrated a greater increase in dynamic expressivity, whereas male students more frequently improved in temporal regularity. These differences may reflect distinct strategies for processing analytical feedback, consistent with prior observations on variability in interpretative approaches across genders (Jia, 2023; Jiang, 2025).

The analysis by piece type indicated that substantially greater gains occurred in works with highly developed phrasing structures—such as Chopin's Prelude Op. 28 No. 4 and the first movement of Beethoven's "Moonlight" Sonata. These pieces are traditionally used to cultivate expressive thinking due to their broad dynamic ranges and dramatic phrasing contours (Li, 2022; Liu & Guo, 2025). In contrast, the Chinese folk melodies, which are more uniform in their dynamic profile, showed modest improvements. This pattern aligns with evidence on the genre-

dependent effectiveness of analytical feedback (Tang, 2025; Zhang et al., 2024).

Despite the significant findings, the study has several limitations. First, the sample consisted exclusively of students from Beijing music colleges, which limits the generalizability of the results to other regions or educational systems. Previous research has noted the cultural specificity of emotional expressiveness development, including differences in pedagogical traditions within China (Cheng et al., 2025; Yin & Guo, 2025). Second, the repertoire consisted of relatively simple works. Although suitable for phrasing analysis, such pieces contain a limited range of timbral, polyphonic, and pedaling challenges, reducing the possibility of a comprehensive assessment of expressivity (Wang & Iida, 2024).

An additional limitation concerns the model's inability to analyze several important performance parameters, including pedaling, articulation, and the acoustic coloration of note onset. These dimensions are critically important in professional interpretation and have repeatedly been identified as central indicators of expressiveness (Rizal & Milyartini, 2024; Yeh, 2016). Their absence in the algorithm constrains the comprehensiveness of the conclusions. Future expansion of the model toward multiparametric analysis may substantially enhance its analytical sensitivity.

Another factor relates to the controlled experimental environment. All lessons were conducted under standardized conditions, with fixed duration and without monitoring the actual amount of individual practice. While such control strengthens internal validity, it reduces external validity, as students in real instructional settings typically engage in less regular and more heterogeneous practice routines (Slavuj et al., 2024; Wang, 2025). A promising direction for future research involves integrating AI-supported feedback into natural pedagogical contexts and conducting longitudinal observations.

Beyond confirming the effectiveness of AI-supported feedback, the present findings suggest that its impact operates through meaningful changes in learners' psychological processing during practice. From a cognitive load perspective, the AI system appears to reduce extraneous load by transforming abstract expressive goals into structured, interpretable indicators. Traditional instruction often requires students to simultaneously integrate verbal explanations, technical execution, and expressive intent, which can strain working memory and hinder expressive learning. By contrast, AI-based feedback externalizes key expressive dimensions (e.g., dynamic shaping, phrasing coherence) into stable reference cues, thereby facilitating more efficient allocation of cognitive resources toward germane processing related to expressive refinement (Han, 2025; Kim et al., 2022; Lei & Liu, 2022). This reduction in cognitive ambiguity may help explain why students were better able to implement expressive adjustments in musically demanding passages.

At the same time, AI-supported feedback appears to enhance metacognitive regulation and attentional resource allocation, both of which are critical for expressive musical performance. Prior research indicates that students often struggle to accurately monitor emotional conveyance without external benchmarks, leading to diffuse or misdirected practice strategies (Bonnaire & González-Moreno, 2022; Liao, 2022). The AI system provides consistent performance-related signals that support self-monitoring and evaluative judgment, enabling learners to align intended and realized expressive outcomes more effectively. In doing so, it also guides attention toward diagnostically relevant expressive features, helping students prioritize global phrasing and emotional coherence over isolated technical details (Hu, 2024; Jia, 2023; Zhang et al., 2024). Importantly, this process does not constrain artistic interpretation but functions as a cognitive scaffold, supporting interpretative autonomy within a hybrid human–AI pedagogical framework (Cheng et al., 2025; Peng, 2024; Wang, 2025).

An important alternative explanation for the observed intervention effects concerns potential differences in students' independent practice time, which was not directly instrumented through objective tracking

tools. Although comparable instructional schedules and lesson durations were implemented across groups, variations in individual practice intensity outside formal lessons could have contributed to changes in expressivity outcomes. This limitation should be considered when interpreting the results, as increased engagement or practice efficiency in one group may partially account for performance gains. At the same time, the absence of systematic evidence suggesting substantial between-group disparities in practice behavior, together with the magnitude and consistency of the observed effects, reduces the likelihood that practice time alone explains the findings.

In sum, the findings suggest that a hybrid “human + AI” model can enhance the quality of interpretive development by combining objective data with human artistic judgment. Unlike purely algorithmic or exclusively pedagogical approaches, this system reduces instructor-related subjectivity and supports the formation of stable skills for reflective control over expressive performance.

The additional sensitivity analysis conducted without applying weighting coefficients to expert ratings showed that the overall pattern of results remained stable: the AI-supported feedback group continued to demonstrate greater gains in expressiveness compared to the control group. However, the effect sizes were somewhat smaller, indicating that listeners' musical experience influences rating precision, while still confirming the robustness of the main findings regardless of the weighting scheme. This suggests that the adopted approach to processing expert evaluations does not determine the outcomes but rather refines their interpretation.

A promising direction for future research involves conducting long-term longitudinal studies. The present intervention spanned a single semester, allowing for the assessment of short-term changes but providing no insight into the durability of the acquired skills. It is essential to examine whether improvements in expressiveness persist over 6–12 months, how students continue to apply analytical feedback after the conclusion of the experiment, and whether delayed effects emerge related to metacognitive development and deeper interpretive thinking. Long-term data would also help determine whether students habituate to AI-generated indicators or, conversely, become more autonomous over time.

Since the study is associated with the development of a digital educational technology, future work must take into account international regulatory requirements—most notably the European Accessibility Act (EAA) and related standards governing the accessibility of digital systems. This includes ensuring that the software is usable for students with diverse cognitive and motor characteristics, as well as adapting the interface to the principles of universal design. Given the growing emphasis on digital inclusivity, integrating accessibility criteria is a necessary condition for the practical implementation of such systems in European and international educational environments.

Finally, the model developed in this study has cross-instrumental potential. Although the present intervention focused on piano performance, the BiLSTM architecture with an attention mechanism can be adapted for analyzing expressiveness in violin, guitar, wind instrument, and even vocal performance, provided that instrument-specific parameters are incorporated (e.g., vibrato, articulation, attack characteristics, pedaling patterns, or breathing profiles). Such cross-instrumental transferability opens the possibility of creating a universal analytical feedback tool capable of supporting interpretive learning across different domains of music education. However, such extensions require separate validation and careful consideration of the unique performance features of each instrument, representing a promising avenue for future research.

Overall, the findings contribute to the broader evidence demonstrating that emotional expressivity can be systematically analyzed and enhanced through AI-supported instruction. Although expressive performance is traditionally viewed as subjective and difficult to formalize (Hu, 2024; Meng, 2024), this study shows that deep learning systems can capture meaningful patterns related to emotional delivery and guide

students toward more expressive interpretations. These results highlight AI's emerging potential to serve not only as an evaluative tool but also as a catalyst for artistic growth within music education.

5. Conclusions

This study demonstrates that the AI-driven DLM-based system effectively enhances emotional expressivity in piano performance. Students who received AI-supported feedback showed considerably higher improvement than those taught through traditional instruction, which confirms the practical value of combining automated analysis with expert-informed assessment.

The findings indicate that AI technologies can serve as a meaningful tool for developing expressive and interpretive skills in music education. The proposed model has potential applications in music schools, universities, and digital learning platforms. It may also be adapted for other musical instruments or used within distance-learning environments. Future research should examine the system's performance with more diverse samples, a more complex repertoire, and multimodal data.

CRediT authorship contribution statement

Xiaodong Deng: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Consent to participate

Informed consent was signed by participants.

Consent for publication

Not applicable.

Ethical approval

The research was conducted ethically in accordance with the World Medical Association Declaration of Helsinki. The research received approval from the Local Ethics Committee of Shenyang Conservatory of Music (Protocol N^o 100,742 dated from September 8, 2024).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares that there are no competing interests.

Data availability

All data generated or analyzed during this study are included in this published article.

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